research grants. The drilling equipment and laboratory facilities were supplied by CSIRO (Mineral Physics) and we are particularly grateful for the assistance and advice of Dr Brian Embleton. The authors have also benefited from discussions with Dr Patrick Coleman and Professor Gordon Packham.

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Analysis of Palaeomagnetic Data from the New Hebrides

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Abstract

Palaeomagnetic samples have been collected on eight islands of the New Hebrides island arc. Data are presented for 31 stable sites out of 46 with ages ranging from Pleistocene to early Late Miocene. These data show 30° clockwise rotation of the arc commencing 6 m.y. BP. Synthetic polar wander paths corresponding to microplate rotation may be computed from known major plate polar wander paths. Thus a physiographically reasonable reconstruction of the New Hebrides plate is shown to have a polar wander path which compares well with the observed data. A reconstruction of a pre-Late Miocene double arc, consisting of the Solomons, New Hebrides-Vityaz, Fiji and Lau-Tonga island arcs, is possible which supports interarc geological correlations and which suggests Plio-Pleistocene growth of the North Fiji Basin by r-r-r *triple junction development concomitant with the development of the Lau Basin.*

Introduction

Gill and Gorton (1973) presented a reconstruction of the New Hebrides island arc prior to 8-10 m.y. BP. These authors rotated the island chain about 45° anticlockwise to align it adjacent to the Vityaz Trench. Prior to that time, subduction was presumed to proceed in a southwesterly direction. The Fiji Platform was also rotated in their reconstruction about 45° clockwise, placing it in line between the New Hebrides arc and the Lau Ridge. This reconstruction was based upon subduction polarity profiles for Fiji and the New Hebrides which suggested both arc reversal (in the case of the New Hebrides) and easterly migration of the Tonga island arc from Pliocene to Recent times. The resultant 70° bend made by the Tonga-Vityaz trench near Fiji roughly follows the present day 625 km earthquake contour. This extreme case of arc rotation is clearly supported by physiography and arc polarity as indicated by a pre-Late-Miocene change in volcanism (Colley and Warden, 1974), but lacks supportive data.

Other authors have suggested less dramatic rotation. Chase (1971) carried out a reconstruction of the New Hebrides and Fiji blocks based upon interpretation of seafloor spreading magnetic anomalies in the North Fiji Basin. The proposed tectonic history was complex. The present day New Hebrides microplate (M) is considered to be quite young. In the interval to 10 m.y. BP the New Hebrides was part of the Pacific plate, and from 10 to 21 m.y. BP it was part of the Indo-Australian plate. Prior to that time it was back on the Pacific plate. The palaeogeography for the 10-21 m.y. BP interval places the

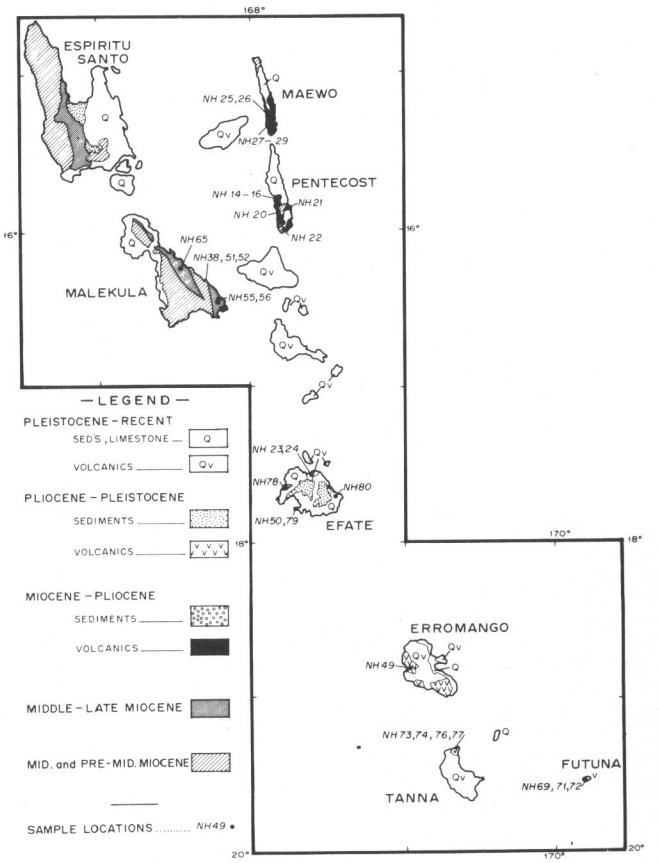


FIGURE 1

Generalised geology of the New Hebrides showing stable Upper Miocene to Pleistocene palaeomagnetic site locations

New Hebrides and Fiji about 400-500 km behind the active southwest-dipping Vityaz subduction zone. This is not obviously in accord with geological evidence presented above. Falvey (1975) presented a simplified interpretation of Chase's ideas (using growth offset transform faults) which brought the New Hebrides closer to the Vityaz Trench. In terms of plate rotation, however, it was still treated as part of the Pacific plate.

Subsequent to that work there had been no more recent progress on the tectonic development of the North Fiji Basin. Thus the simple question remained: was the New Hebrides island arc part of the Pacific plate (with its predictable polar wander path) or was it part of a more dramatically rotating microplate? The results reported here represent part of a comprehensive palaeomagnetic field sampling programme of 80 sites undertaken in 1976-77 to answer that question.

Field sampling and data reduction

The data presented here are from 46 sites in rocks younger than Middle Miocene. The regional geology and site locations of the 31 stable sites (see below) are shown in Figure 1. All sites were cored with a motorised rock drill; about 5 or 6 unbroken cores per site. *In situ* orientation was by magnetic and sun compasses. Where possible, structural variability was chosen to facilitate fold tests. However, imprecise or unrecognised structure remains a major problem in these islands. No contact tests were possible.

Specimen analysis was carried out on a spinner magnetometer at Mineral Physics laboratory, Commonwealth Scientific and Industrial Research Organisation (CSIRO), North Ryde. Where necessary, or possible, both thermal and alternating field demagnetisation was carried out on specimens from a single core. A stable and useful site was recognised if:

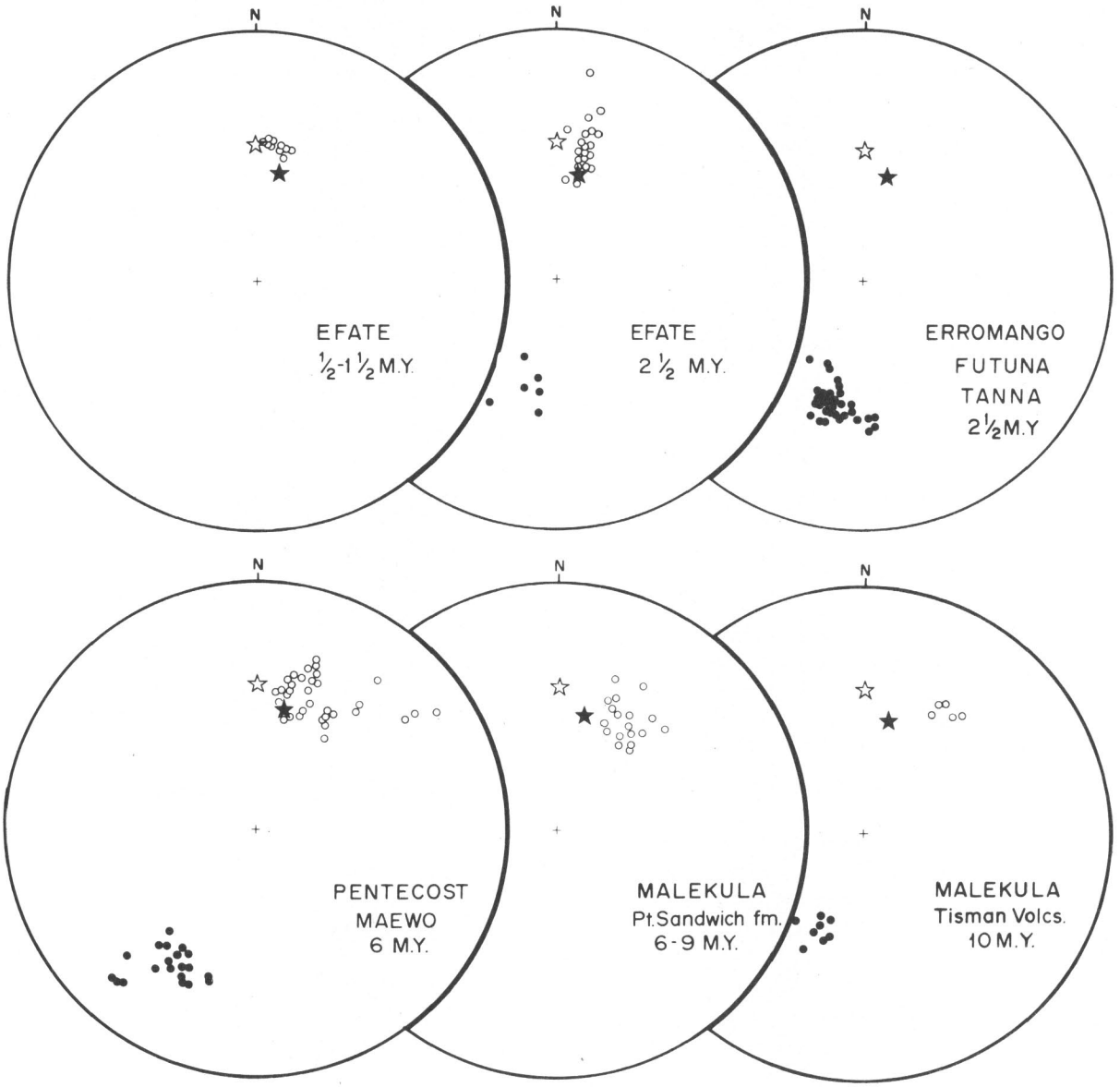


FIGURE 2

Directions of magnetisation in all stable specimens in all stable sites given in Table 1.

Mean site NRM data – New Hebrides

Site No.	Rock Description	Age	No Cores	Site average palaeodirection (after structural correction)		A_{95}	Sample (core) average pole postition		A_{95}
				Decl.	Incl.		Lat.	Long.	
NH 23	Quoin Hill Volcs. (flow)	Pleis. (Early?)	3	13.0	–34.1	3.6°	–82.1	80.8	2.5°
24	" " " (dyke)	"	7	6.3	–31.9	2.1°			
49	Plateau Fm. (flow)	2.4 m.y.	5	200.0	36.2	3.7°			
69	Futuna Volcs. (dyke)	1.8–2.4 m.y.	4	195.6	37.7	2.8°			
71	" " "	"	5	183.6	19.3	4.7°			
72	" " (flow)	"	3	198.1	37.6	3.7°	–78.5	92.2	2.8°
73	Green Hill Volcs. (flow)	2.4 m.y.	5	194.6	33.2	4.0°			
74	" " " "	"	5	190.3	43.9	3.5°			
76	" " " "	"	5	186.7	35.1	3.3°			
77	" " " (dyke)	"	4	204.8	48.8	8.2°			
50	Rentabau Tuffs	Late Plio.	8	11.5	–29.8	6.1°			
78	" "	" "	6	191.4	44.5*	6.5°			
79	" "	" "	7	14.1	–39.7	4.6°	–77.6	89.0	3.4°
80	" "	" "	6	14.9	–28.6	6.6°			
14	Pentecost Gp:								
	Lemulmu Fm. (flow)	6.0 m.y.	3	221.4	11.2	2.2°			
15	" " "	"	4	209.5	23.3	4.7°			
16	" " "	"	6	217.4	27.9	6.1°			
20	Dyke in Olambe Fm.	5.6–6.1 m.y.	7	203.5	25.9	4.5°			
21	Pentecost Gp:								
	Wari Fm. (flow)	5 4–6.2 m.y.	6	20.8	–26.0	8.4°			
22	Pentecost Gp:								
	Warbot Fm. (flow)	6.4 m.y.	3	54.7	–12.3	6.7°	–64.2	80.6	3.1°
25	Sth. Maewo Volcs. (dyke)	~5½ m.y. (?)	5	31.5	–34.2	2.2°			
26	" " " "	"	4	12.6	–38.0	2.8°			
27	" " " (flow)	"	4	34.9	–23.5	11.5°			
28	" " " "	"	7	14.6	–32.2	4.9°			
29	" " " (dyke)	"	5	15.3	–25.7	5.2°			
55	Pt Sandwich Fm.	Late Miocene	6	35.5	–38.0	7.5°			
56	" " "	" "	6	24.0	–27.8	5.1°	–59.4	87.6	3.8°
65	" " "	" "	6	36.7	–36.7	5.7°			
38	Tisman Volcs. (dyke)	10.4–10.5 m.y.	4	206.5	43.4	10.0°			
51	" " "	"	5	33.1	–28.4	4.2°	–62.5	92.8	5.3°
52	" " "	"	4	203.4	45.9	6.0°			

* Both stable normal and reversed polarity samples within site.

1. demagnetisation intensities were not abnormal;
2. a definite "best grouping" of magnetisation directions could be recognised during demagnetisation (usually with $A_{95} < 8°$); and
3. palaeofield inclinations after structural correction were not vastly different from those possible for the Pacific plate or New Hebrides microplate at that latitude over 10 m.y.

Magnetisation directions corresponding to the best group of all specimens in all stable sites are given in Figure 2 and site data are listed on Table 1.

Results and interpretation

a. Efate (½-1½ m.y. BP): These are Pleistocene volcanics with good structural correction and

which show a small but significant difference in declination from the dipole field direction.

b. Efate (2½ m.y. BP): These Upper Pliocene fine-grained tuffs show a small improvement in site grouping after structural correction and 12° declination east of the dipole field.

c. Erromango, Futuna, Tanna (2½ m.y. BP): Most are well dated volcanics just younger than the Matuyama/Gauss reversal (2.41 m.y. BP). The improvement of site grouping upon structural correction is just below the 95% significance level as defined by McElhinny (1964). The data also show 12° declination east of the dipole field.

d. Pentecost, Maewo (6 m.y. BP): These are well dated, latest Late Miocene volcanics show an improvement on site groupings upon structural correction significant at better than the 99% level as defined by McElhinny (1964). The declination of 29° east of the dipole field clearly shows significant microplate rotation.

e. Malekula (Port Sandwich Formation): These are poorly dated and may encompass both Late and Middle Miocene. The author believes them to be younger than the 10.5 m.y. BP Tisman Volcanics. There is no significant change in declination east of the Maewo/Pentecost data.

f. Malekula (Tisman Volcanics): These sites were well dated by Gorton (1974) at 10.5 m.y. Again the declination is the same as in the younger Maewo and Pentecost rocks. The Malekula data clearly show that the inferred microplate rotation must have commenced about 6 m.y. BP.

The apparent south pole positions shown on Figure 3 and listed in Table 1 were computed from all the best group specimen directions in all stable sites. The apparent pole position for sites in the age range 6 to 10 m.y. BP is covered by a region of overlapping circles of confidence at about 62°S and 83°E. The inferred initiation of arc rotation is younger than all previous estimates (Gill and Gorton, 1973; Falvey, 1975; Carney and MacFarlane, 1977).

The apparent pole position for the 4-4½ m.y. BP data from Viti Levu presented by James and Falvey (*this issue*) shows an opposite but comparable microplate rotation in an adjacent and comparable tectonic setting. Gill and McDougall (1973) suggested that between 5 and 6 m.y. BP the site of

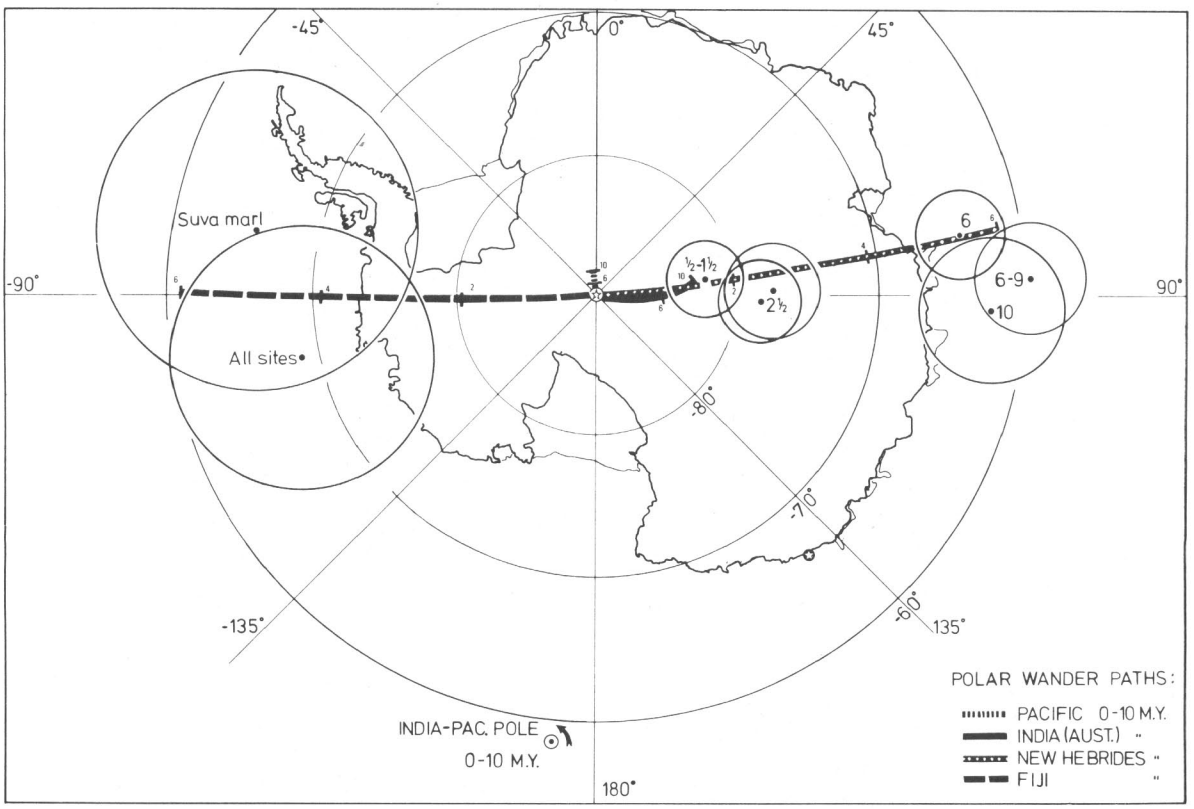


FIGURE 3

Apparent pole positions for the New Hebrides (eastern hemisphere date) with ages in millions of years. The apparent pole positions for Fiji (western hemisphere data) is from James and Falvey (*this issue*). The polar wander path for Australia is taken from McElhinny *et al.* (1974) and that for the Pacific has been computed using the Indo-Australian/Pacific stage pole (see text). The other polar wander paths are synthetic and have been computed using the reconstruction poles used to generate Figure 4.

subduction changed near Fiji. These observations taken together suggest that arc rotation (in the opposite sense) commenced in Fiji at the same time as in the New Hebrides – namely 6 m.y. BP.

Miocene arc reconstruction

It is clear from the apparent and inferred Plio-Miocene pole positions of the New Hebrides and Fiji that significant arc rotation has occurred. This rotation can be seen relative to the Australian Plio-Miocene polar wander path (Figure 3) taken from data in McElhinny *et al.* (1974). The Pacific polar wander path was computed using the India-Pacific plate stage pole computed by Packham and Falvey (1976) for that interval (0-10 m.y. BP).

However, a unique microplate reconstruction is not possible using apparent pole positions data alone. Any finite rotation may be composed of two successive rotations and one of those arbitrarily be about the geographic pole itself. This gives rise to an uncertainty in longitude which may only be resolved using physiographic or auxiliary geotectonic data. In this case the relative motion between the Indo/Australian and Pacific plates is well known to 10 m.y. BP (*op. cit.*). Micro-plates between these major plates are somewhat limited in the extent to which their longitude is undeterminate (at least where the major plate boundary is not east-west). Indeed if the present easterly-dipping subduction is assumed to be 6 m.y. old, and island overlap is not permitted with the Vityaz "arc" (Horizon, Pandora, and Charlotte Banks, and Mitre and Anuda Islands), then the reconstruction of the New Hebrides island arc is very well constrained.

It is fortunate that after reconstruction of the New Hebrides with respect to the Vityaz arc and then with respect to the Lau Ridge, only a relatively restricted physiographic 'space' remains. The resultant rotation of the Fiji Platform into a location within the arc is thus also well constrained. To complete the arc, it is necessary only to reconstruct the Tonga arc with respect to the Lau Ridge (Weissel, 1977; James and Falvey, *this issue*).

Figure 4 shows a computer-modelled reconstruction using the above constraints. In addition, the Solomon and Vityaz arcs have been rotated on the Pacific plate clockwise with respect to the Australian plate about the present Indo-Australian/Pacific stage pole (Packham and Falvey, 1976). The Lau Ridge has been left on the 'stationary' Australian plate. The rotation of the New Hebrides with respect to the Australian plate is 24.4° about 6.1°N, 169.0°E. The rotation of the Fiji Platform with respect to the Australian plate is – 35.5° about 18.8°S, 175.3°W. The resultant synthetic polar wander paths for the New Hebrides and Fiji are shown in Figure 3. The agreement is considered very good given that the Fiji pole data is relatively poor (James and Falvey, *this issue*).

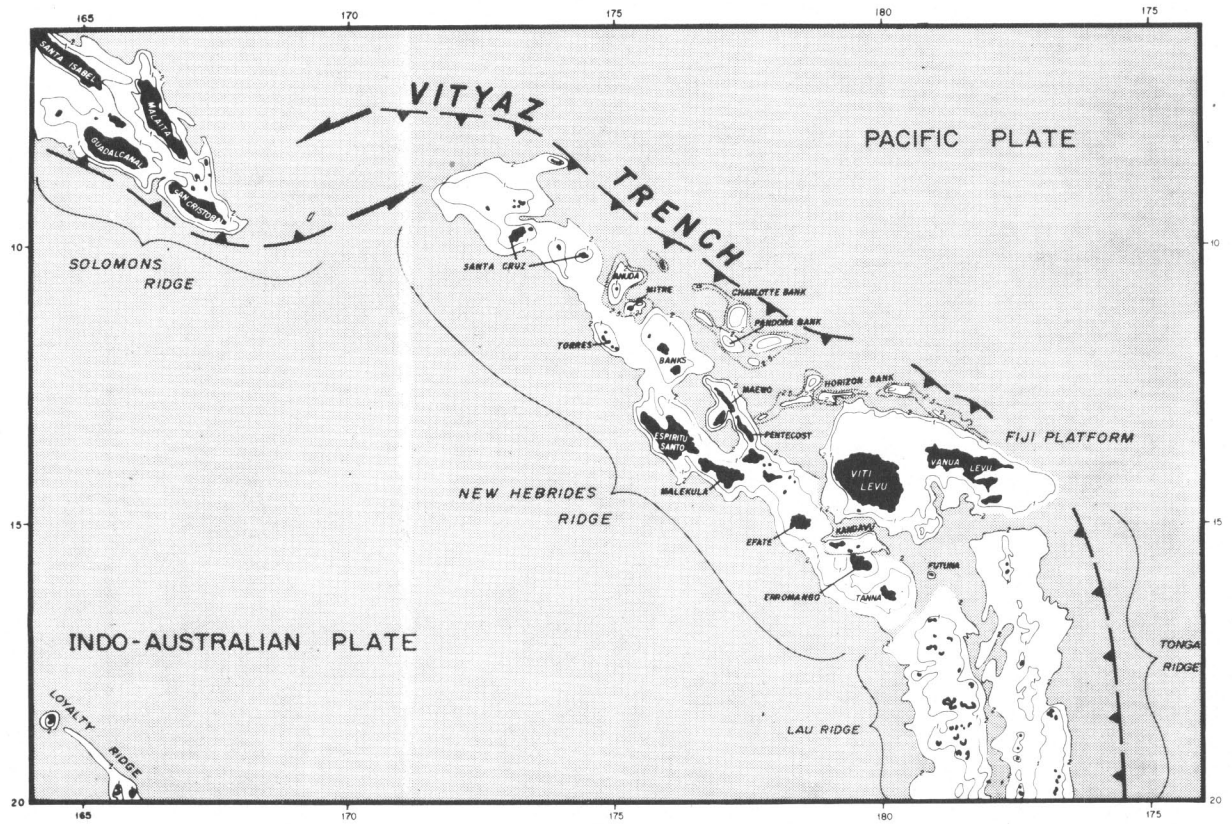


FIGURE 4

Proposed 6 m.y. BP reconstruction of the Melanesian Arc. The Tonga-Lau reconstruction is after Weissel (1977). The reference frame is fixed to the Indo-Australian Plate. The Plio-Pleistocene Pacific/Indo-Australian convergence vector is shown by the transform offset implied between the Solomon and New Hebrides arcs. It is possible that arc reversal in the Solomons occurred about 4 m.y. earlier (10 m.y. BP).

Significance of the reconstruction

If the reconstruction poles represent relatively stationary and constant microplate/plate rotation poles, then general inferences may be made concerning the plate boundary geometry in the region during the Plio-Pleistocene. Figure 5 represents an interpretive pattern of plate boundaries responsible for the North Fiji and Lau basins during this period. Note that the Fiji/Tonga/Pacific triple junction is unstable at 6.m.y. BP and evolves into two stable triple junctions, encompassing the 'Peggy' plate. The latter was envisaged by Weissel (1977).

The evolutionary scheme proposed in Figure 5 justified the pre-Upper Miocene geological correlations between Fiji and the New Hebrides noted by Carney and Macfarlane (*this issue*). It is probable that this extensive arc system has existed at least since the formation of the South Fiji Basin in the

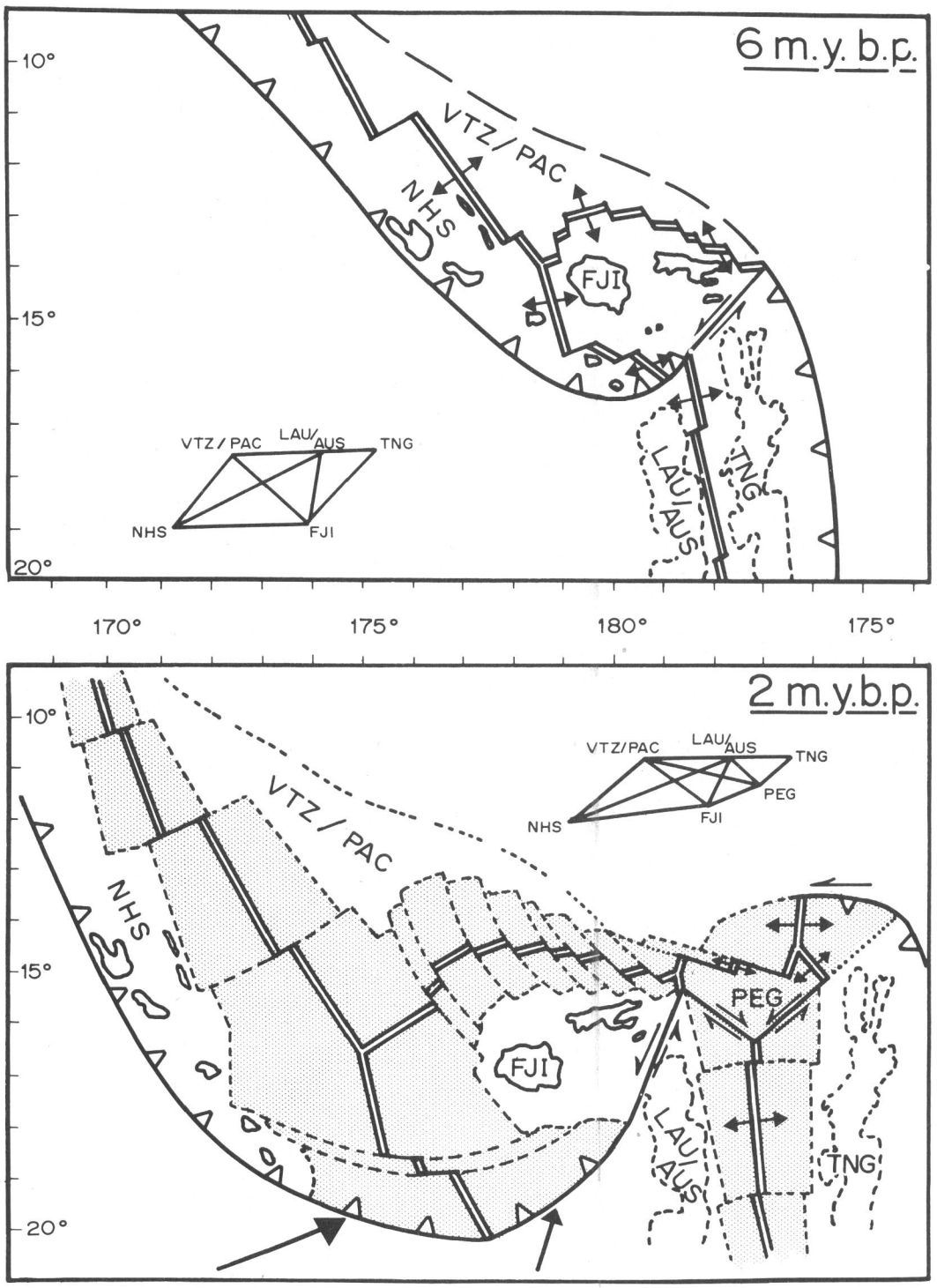


FIGURE 5

Possible pattern of seafloor growth within the North Fiji and Lau Basins based on the reconstruction in Figure 4. The insets show the generalised plate velocity vectors.

Oligocene. However, the cause of the earlier tectonism in Fiji and the New Hebrides has not been explained.

The history of Pliocene to Recent volcanism in Fiji (Gill and Gorton, 1973) and in the New Hebrides (Colley and Warden, 1974) can be adequately explained by the scheme outlined in Figure 5. The present day distribution of earthquakes (Chase, 1971) does not correlate well with this model extended to the present day. The possibility that very recent plate kinematic changes have occurred, as suggested by Chase, cannot be excluded.

Acknowledgements

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Lower to Middle Miocene Sediments on Maewo, New Hebrides, and their Relevance to the Development of the Outer Melanesian Arc System

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Abstract

Lower to Middle Miocene sediments on Maewo contain clasts derived from the 'Vitiaz arc', a former tholeiitic volcanic belt similar in age, lithology and geochemistry to Upper Eocene-Middle Miocene rocks on Viti-Levu, Fiji. The configuration of Outer Melanesia from the New Hebrides through Fiji to Tonga-Lau in the Early to Middle Miocene was a double arc compising a frontal arc (Vitiaz arc — Viti Levu — Tonga Ridge) and a rear arc (Western Belt-Lau Ridge) couple. Tectonism in the Middle Miocene brought about inter-arc rifting between the Vitiaz arc-Western Belt couple and the anti-clockwise rotation of Viti Levu into a rear arc position; further fragmentation and the development of marginal basins had also affected the Tonga-Lau couple by the Early Pliocene. Mio-Pliocene lavas of the New Hebrides Eastern Belt were erupted in the Vitiaz arc-Western Belt inter-arc basin in response to further rifting and crustal thinning associated with the developing proto-North Fiji Basin. Expansion of this basin was accompanied by subduction reversal from the eastern to western margins of the New Hebrides Ridge which then migrated south-westwards to its present position. The present configuration of the New Hebrides Ridge is atypical of that usually attributed to SW Pacific arc systems.

Introduction

The New Hebrides forms part of the Outer Melanesian island arc system, a disconnected series of island chains considered by many workers as a once-continuous belt of volcanism located along the outer margin of the Indo-Australian Plate and linking the New Hebrides to the Solomon Islands, Fiji and the Tonga-Lau Group. Fragmentation of this volcanic belt began in the Middle Miocene and was accelerated in Mio-Pliocene times by a change in convergence between the Australian and Pacific plates (e.g. Chase, 1971; Packham, 1973), one result being a transfer of trench systems from the eastern to the western margin of the New Hebrides arc and the reversal in the direction of subduction. The New Hebrides subsequently underwent clockwise southwesterly rotation and a marginal sea, the North Fiji Basin, was created behind the arc (Chase, 1971; Falvey, 1975, *this issue*) (Figure 1).

Arc fragmentation has obscured detailed geological correlations between the SW Pacific islands as they now exist, and there is little in the way of field evidence to aid reconstruction of the region prior to its break-up. One of the most extensive discontinuities is that between the New Hebrides